CIS 344 Group Project
Library Management System
Course: CIS 344 – Database Design and Programming
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Project: Library Management System
Group Members

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1. Project Overview

The Library Management System is a web-based application designed to organize and manage basic library operations. The system allows library staff to manage members, books, borrowing transactions, and book returns. Library members can create an account, log in, view available books, and review their borrowing information.

The application was developed using PHP for server-side processing, MySQL for database management, and HTML, CSS, and JavaScript for the user interface.

The project demonstrates how a relational database can be connected to a web application to store, retrieve, and update information efficiently.

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2. Project Scope

The scope of the Library Management System includes the management of:

- Library members
- Books
- Book availability
- Borrowing transactions
- Loan records
- Due dates
- Book returns
- Member accounts
- Administrative access

The system contains separate functionality for library members and library staff/administrators. Members can create accounts, log in, view available books, and view their borrowing history. Administrators can manage members and books, create borrowing transactions, process returns, and view active loan information.

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3. Project Objectives

The main objectives of the project are to:

1. Design a relational MySQL database for managing library information.
2. Create relationships between members, books, and loans using primary and foreign keys.
3. Develop a PHP backend that communicates with the MySQL database.
4. Create an organized and responsive user interface using HTML, CSS, and JavaScript.
5. Allow administrators to add members and books.
6. Allow borrowing and returning of books.
7. Track the number of available copies of each book.
8. Allow members to create accounts and access their library information.
9. Use SQL JOIN queries to retrieve information from multiple related tables.
10. Apply input validation and basic security practices.

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4. Database Design

The database is named:

library_management_system

The main database tables are:

MEMBERS

The MEMBERS table stores information about registered library members.

Important attributes include:

- member_id – Primary Key
- full_name
- email
- phone
- password
- created_at

The member_id uniquely identifies each library member.

BOOKS

The BOOKS table stores information about books available through the library.

Important attributes include:

- book_id – Primary Key
- title
- author
- isbn
- total_copies
- available_copies
- created_at

The book_id uniquely identifies each book.

The system tracks both the total number of copies and the number of copies currently available.

LOANS
The LOANS table records borrowing transactions.
Important attributes include:

- loan_id – Primary Key
- member_id – Foreign Key
- book_id – Foreign Key
- borrowed_at
- due_date
- returned_at

The LOANS table connects MEMBERS and BOOKS.
A member can have multiple loan records, and a book can appear in multiple loan records over time.

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5. Database Relationships
The database contains the following primary relationships:
MEMBERS 1 LOANS
One member can have many loan records, while each loan belongs to one member.
BOOKS 1 LOANS
One book can have many loan records over time, while each loan refers to one book.
The LOANS table therefore acts as the transaction table connecting MEMBERS and BOOKS.
Foreign keys are used to maintain these relationships and help preserve referential integrity.

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6. SQL JOIN Implementation
SQL JOINs are an important part of the Library Management System because information about a borrowing transaction is stored across multiple tables.
For example, the system uses the MEMBERS, LOANS, and BOOKS tables together to display information about active loans.
A three-table JOIN can retrieve the member's name, book title, borrowing date, and due date:

```
None
SELECT
    m.full_name,
    b.title,
    l.borrowed_at,
    l.due_date
FROM LOANS l
```

```
JOIN MEMBERS m
   ON l.member_id = m.member_id
JOIN BOOKS b
   ON l.book_id = b.book_id
WHERE l.returned_at IS NULL;
```

This query demonstrates how relational database relationships can be used to combine information stored in different tables.

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7. Application Development

The application was developed using:

Frontend

- HTML
- CSS
- JavaScript

Backend

- PHP

Database

- MySQL / MariaDB
- MySQL Workbench

The PHP backend processes forms and communicates with the database. The application can add members and books, create borrowing transactions, process book returns, and retrieve information from the database.

The frontend provides navigation and interfaces for viewing and managing library information.

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8. Group Member Contributions

Afia Boateng

Afia worked primarily on the database portion of the project. Her responsibilities included:

- Designing the database structure.
- Creating the MEMBERS, BOOKS, and LOANS tables.
- Defining primary keys and foreign keys.
- Developing and testing SQL JOIN queries.
- Testing database relationships.
- Working on the ER/EER database design.
- Contributing to the database design section of the final report.

AbdulMateen Mudasiru

AbdulMateen worked primarily on the PHP backend. His responsibilities included:

- Developing PHP backend functionality.
- Creating forms for adding members and books.
- Implementing borrowing and returning functionality.
- Connecting PHP to the MySQL database.
- Implementing input validation and security checks.
- Testing and troubleshooting PHP functionality.

Ashraf Abubakar

Ashraf worked primarily on the frontend and documentation. His responsibilities included:

- Developing the frontend using HTML, CSS, and JavaScript.
- Designing application navigation.
- Creating interfaces for books, members, loans, borrowing, and returns.
- Improving the organization and responsiveness of the interface.
- Contributing to the user guide.
- Preparing screenshots and demo materials.
- Helping organize the GitHub repository and README.

All group members contributed to testing the complete application, reviewing database queries, preparing project documentation, and preparing the final demonstration.

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9. Project Files

The project contains the following major files and resources:

- Library_Management_System.mwb – MySQL Workbench database model
- Library_Management_System.sql – SQL database script
- PHP application files
- HTML/CSS/JavaScript frontend files
- README documentation
- Database diagrams
- User guide and screenshots
- Final project report

The project's source code is maintained in the group's public GitHub repository.

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10. Testing

The database and web application were tested to verify that:

- Members can be stored successfully.
- Books can be stored successfully.
- Primary and foreign key relationships function correctly.
- Books can be borrowed.
- Books can be returned.
- Available book quantities are updated.

- Active loans can be retrieved.
- SQL JOIN queries correctly combine related information.
- Member and administrator pages connect correctly to the database.
- User input is validated before database operations are performed.

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11. Challenges

During development, the group encountered several technical challenges, including database connection configuration, integrating the frontend with the PHP backend, and ensuring that the application communicated with the correct MySQL database.

These issues were addressed through configuration testing, database connection troubleshooting, and application testing.

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12. Conclusion

The Library Management System demonstrates the development of a database-driven web application using MySQL, PHP, HTML, CSS, and JavaScript.

The project uses relational database concepts such as primary keys, foreign keys, one-to-many relationships, constraints, and SQL JOINs. The completed system provides functionality for managing members, books, loans, borrowing, and returns.

The project also demonstrates how frontend, backend, and database technologies can work together to create a functional information system.